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ASTRONOMY AND ASTROPHYSICS

Formation of primordial molecules and thermal balance in the early Universe

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Abstract. The role played by molecular radiative processes after the decoupling of matter and radiation is discussed. In the physical environment of the expanding Universe, molecules such as H_2, HD and LiH can be formed and contribute to the thermal evolution of the medium.

In this paper, a detailed description of the primordial molecular chemistry is presented. The molecular cooling and heating functions have a slight influence on the decrease of the temperature of the matter.

The diffuse radiative background due to radiative transitions of the primordial molecules is also calculated and found to be far below observational detection threshold.

Key words: cosmology – atomic and molecular processes

1. Introduction

After recombination, although the density-decrease acts against molecular formation, it turns out that the temperature is small enough for this formation to occur. The Universe expansion time scale is comparable to some of the reaction time scales and a time-dependent analysis must be performed to describe the evolution of molecular abundances. The importance of radiative molecular processes in thermal exchanges is well known in the interstellar medium, and molecules in the cooling Universe could modify the conventional picture of the evolution of the matter temperature, for example during the formation of density condensations (particularly in the initial conditions suitable to gravitational collapse and galaxy formation).

We propose here to examine whether formation of molecules in the domain of redshifts $z < 1000$ can affect the temperature of matter. Peebles (1968) has shown that after recombination, owing to Compton scattering the temperature of the matter do not follow the adiabatic expansion. In a more detailed study, Jones & Wyse (1985) have given an analytical approximation of the ionisation equation of the primeval plasma. To our knowledge, the thermal role of the molecules has been investigated only for collapsing objects (Lepp & Schull 1984, Hutchins 1976) with density and temperature ranges orders of magnitude greater than those considered in this paper.

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In Sect. 2, we present the time dependent primordial chemistry after recombination, and in Sect. 3, the molecular cooling and heating function induced by primordial molecules. In Sect. 4, the contribution of chemical processes on the evolution of the mean temperature of matter is described. In Sect. 5, we discuss the thermal influence of molecular radiative and collisional processes and a comparison is made with the Compton contribution. Finally, we examine the contribution of the molecular radiative processes to the observed CMBR (Cosmic microwave background radiation).

2. Primordial chemistry

The chemical species available from primordial nucleosynthesis are H, H^+, D, D^+, He, and Li, forming molecules such as H_2, HD, LiH and HeH^+. Since the work of Saslaw & Zipoy (1967), the formation of molecules in the primordial gas has been studied by several authors (e.g. Shchekinov & Entél 1983 and Lepp & Shull 1984). This subject has recently been reviewed by Dalgarno & Lepp (1987) and Black (1988). As a result of the expansion of the Universe, the recombination time of hydrogen becomes greater than the Hubble time for $z \lesssim 800$ and the ionized fraction of atomic hydrogen remains constant $\left(\frac{H^+}{H} \sim 10^{-4}\right)$ and above the equilibrium value. The chemical reactions are listed in the Appendix.

Different routes lead to H_2 formation, each initiated by a slow radiative process:

$$H + e^- \mapsto H^- + h\nu \qquad \text{(reaction 1)}$$

$$H^- + H \mapsto H_2 + e^- \qquad \text{(reaction 2)}$$

and

$$H^+ + H \mapsto H_2^+ + h\nu \qquad \text{(reaction 3)}$$

$$H_2^+ + H \mapsto H_2 + H^+ \qquad \text{(reaction 4)}$$

The photodetachment of H^- and the photodissociation of H_2^+ by the background radiation field restrict the abundance of molecular hydrogen formed at the early stages. These radiative processes are calculated using the bound-free photodetachment cross section of Wishart (1979) for H^-, and the photodissociation cross section of Von Busch & Dunn (1972) for H_2^+ and HD^+. Note that most of the dissociation of H_2^+ comes from excited vibrational

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levels: the photon density, higher at short frequencies, compensates the lower population of these levels. In this study, the photodestruction of molecular hydrogen is found negligible using the approximation given by Latter (1989) and Latter & Black (1991).

Although similar processes contribute to the formation of HD, its formation proceeds mainly through:

$$H^+ + D \mapsto D^+ + H \qquad \text{(reaction 13)}$$

$$D^+ + H_2 \mapsto H^+ + HD. \qquad \text{(reaction 15)}$$

We treat photodissociation of HD similarly to H_2 (Latter 1989, Latter & Black 1991). LiH is supposed to be formed through radiative association (Lepp & Shull 1984) and destroyed by photodissociation (Kirby & Dalgarno 1978).

The numerical integration of the coupled chemical equations is an initial value problem for stiff differential equations. We have used a code written by Hindmarsh (1974), which is based upon a Gear (1971) method. Since the reaction rates depend on temperature, the temperature and density evolution equation must be solved simultaneously, which needs in turn the simultaneous determination of molecular cooling and heating rates.

3. Molecular contribution to cooling and heating

Below 3000 K, only the rotational levels of the molecules can be excited (the first vibrational energy level of H_2 is excited for $T \gtrsim 6000$ K).

The population of the rotational levels is mainly due to collisional excitation and deexcitation with H, H_2 and He on one hand and to radiative processes (absorption from CMBR and spontaneous or induced emission) on the other hand. These processes are fast enough to reach a statistical equilibrium between the energy levels, in which case the population of all levels can be computed using detailed balance equations:

$$n_i \sum_{j'<i} (k^X_{ij'} n_X + A_{ij'} + B_{ij'} u^{ij'}) + n_i \sum_{j'<i} (k^X_{ij'} n_X + B_{ij'} u^{ij'}) = \sum_{j>i} (k^X_{ji} n_X + A_{ji} + B_{ij} u^{ij}) n_j + \sum_{j>i} (k^X_{ji} n_X + B_{ji} u^{ji}) n_j \qquad (1)$$

where i, j or j' denote the rotational levels of the molecule; n_i is the abundance of the level i, k^X_{ij} is the rate of collision with species X leading to a transition from level i to level j, A_{ij} the radiative transition probability of spontaneous emission, B_{ij} the Einstein coefficient of induced emission and u^{ij} the intensity of the radiation field at frequency ν_{ij}. The population of the rotational levels, which is related to the mean temperature, is computed for the first 15 rotational levels ($J = 0$ to 14) for H_2, HD and LiH between $z = 1000$ to $z = 5$.

Molecular cooling corresponds to collisional excitation followed by radiative transition, provided that no further absorption occur. In a dust free gas, radiative attenuation is mainly due to self-absorption in the line. The optical depth at the center of a line is given by:

$$\tau_D = \sigma_{ij} N_i \text{ with } \sigma_{ij} = \frac{1}{8\pi\sqrt{\pi}} \frac{\lambda^2_{ij}}{\Delta\nu_D} (A_{ji} + B_{ji} u^{ji})$$

where τ_D is the optical depth at the line center, N_i the total column density of the lower level, σ_{ij} the absorption cross section from level i to level j, λ_{ij} the wavelength of the transition, $\Delta\nu_D$ the Doppler width of the line, A_{ji} the Einstein emission probability.

We obtain an optical depth unity for a mean free path of $L = \frac{1}{n_i \sigma_{ij}}$, where n_i is the population of level i (with typical values of $\sigma_{ij} \sim 10^{-23}\,\text{cm}^2$ and $n_i \sim 10^{-8}\,\text{cm}^{-3}$, we get $L \sim 1000$ M pc). In the specific case of primordial gas, the global expansion of the Universe leads to a significant Doppler shift of the lines at a scale smaller than L. The net effect is that an infrared photon is not absorbed and this leads to an energy loss for matter. Molecular heating is due to radiative excitation from CMBR followed by collisional de-excitation. Notice that, although the radiative de-excitation is orders of magnitude faster than the collisional one (as the excitation rates have the opposite ordering), the full cooling and heating processes must be evaluated.

We introduce the de-excitation collisional probability P^c_{ji}

$$P^c_{ji} = \frac{n_X n_j k^X_{ji}}{n_X n_j k^X_{ji} + n_j A_{ji} + n_j B_{ji} u^{ji}} \qquad (2)$$

and the de-excitation radiative probability P^r_{ji} between the levels i, j.

$$P^r_{ji} = \frac{n_X n_j A_{ji} + n_j B_{ji} u^{ji}}{n_X nj k^X_{ji} + n_j A_{ji} + n_j B_{ji} u^{ji}} \qquad (3)$$

The energy radiated per unit volume is then:

$$\Lambda_{\text{mol}} = \sum_j n_j \sum_i n_X n_i C_{ij} P^r_{ji} \varepsilon_{ji} \qquad (4)$$

and the energy gain from CMBR per unit volume:

$$\Gamma_{\text{mol}} = \sum_j n_j \sum_i n_i B_{ij} u^{ij} P^c_{ji} \varepsilon_{ji}, \qquad (5)$$

where ε_{ji} is the energy difference between levels i and j. The ratio between the cooling and heating is:

$$\gamma = \frac{\Lambda^j_{\text{mol}}}{\Gamma^j_{\text{mol}}} = \exp\left[T_j\left(\frac{1}{T_m} - \frac{1}{T_r}\right)\right] \qquad (6)$$

where the temperature T_j corresponds to the energy ε_{ji}. In the cosmological context after the decoupling between matter and radiation, the radiation temperature T_r is always greater than matter temperature T_m. Thus, the molecular cooling is lower than the molecular heating, and molecules contribute to transfer energy from CMBR to matter, as do Compton interaction.

The relations (4) and (5) are used to estimate the cooling and heating fluxes due to the molecules H_2, H and LiH. Since H_2 is an homonuclear molecule, radiative transitions within the lowest electronic state involve only quadrupole transitions whose rates are taken from Danby et al. (1987). Collisions rates of the H_2 molecule with H, H_2, He, H^+ are the ones used by Abgrall et al. (1992).

HD and LiH molecules have a lower excitation temperature than H_2 (112 K and 21 K respectively). Since both molecules have a dipole moment ($\mu_{HD} = 8.3\ 10^{-4}$ debye, Abgrall et al. 1982 and $\mu_{LiH} = 5.88$ debye, Wharton et al. 1960), radiative decay is much faster. We take $10^{-9}\,\text{cm}^{-3}\,\text{s}^{-1}$ as conventional value for the collision rates of LiH and HD with H.

4. Temperature and density evolution

The adiabatic expansion of radiation is given by:

$$\frac{dT_r}{dt} = -T_r H_0 (1+z)^{3/2} \quad \text{for } \Omega_0 = 1 \qquad (7)$$

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where T_r is the temperature of the radiation, H_0 the Hubble constant, and Ω_0 the density parameter. As first shown by Peebles (1968), coupling between matter and radiation via the Thomson diffusion, after the phase of recombination ($z \lesssim 1000$) in the expanding Universe, leads to a thermal evolution equation:

$$\frac{dT_m}{dt} = -2T_m H_0(1+z)\sqrt{1+\Omega_0 z} + \frac{8\sigma_T a_{bb} T_r^4}{3m_e c} x_e(T_r - T_m), \quad (8)$$

where T_m is the temperature of matter, a_{bb} is the blackbody constant, σ_T the Thomson cross section, x_e the ionisation fraction, m_e the electron mass and c the speed of light (we shall keep further below the radiation temperature equation, Eq. (7), unchanged since the amount of energy transferred from radiation to matter is found to be small). Once the molecules form, the thermal evolution equation, Eq. (10), must also take into account a molecular source term. Due to the expansion and to the chemical reactions, the comoving particle density evolves as:

$$\frac{1}{n}\frac{dn}{dt} = -3H_0(1+z)\sqrt{1+\Omega_0 z} + \frac{1}{n}\left(\frac{dn}{dt}\right)_{ch}, \quad (9)$$

(the subscript ch denotes the contribution of chemical reactions). We thus obtain:

$$\frac{dT_m}{dt} = -2T_m H_0(1+z)\sqrt{1+\Omega_0 z} + \frac{8\sigma_T a_{bb} T_r^4}{3m_e c} x_e(T_r - T_m) + \frac{2(\Gamma_{mol.} - \Lambda_{mol.})}{3nk} + \frac{2\Theta_{ch}}{3nk} - \frac{T_m}{n}\left(\frac{dn}{dt}\right)_{ch}, \quad (10)$$

where $\Gamma_{mol} - \Lambda_{mol}$ is the energy heat-loss function given in Sect. 3. The term $\left(\frac{dn}{dt}\right)_{ch}$ characterizes the variation of density due to chemistry. Θ_{ch} is the chemical heating and cooling:

$$\Theta_{ch} = \sum_i k_i n_i^X n_i^Y \Delta H_i,$$

where k_i is reaction rate of the reaction i (see Appendix), n_i^X and n_i^Y the abundances of species X and Y produced in reaction i and ΔH_i is the enthalpy of the reaction i. These two chemicals contributions remain negligible.

Table 1. Parameters of models A, B. D/H and Li/H are the relative elemental abundance of deuterium and lithium

	Ω_0	Ω_b	D/H	Li/H
Model A	1	0.1	$1.7\ 10^{-5}$ (a)	10^{-10} (a)
Model B	1	1	10^{-4} (b)	10^{-8} (c)

References: (a) Yang et al. (1984), (b) Sale & Mathews (1986), (c) Mathews et al. (1990).

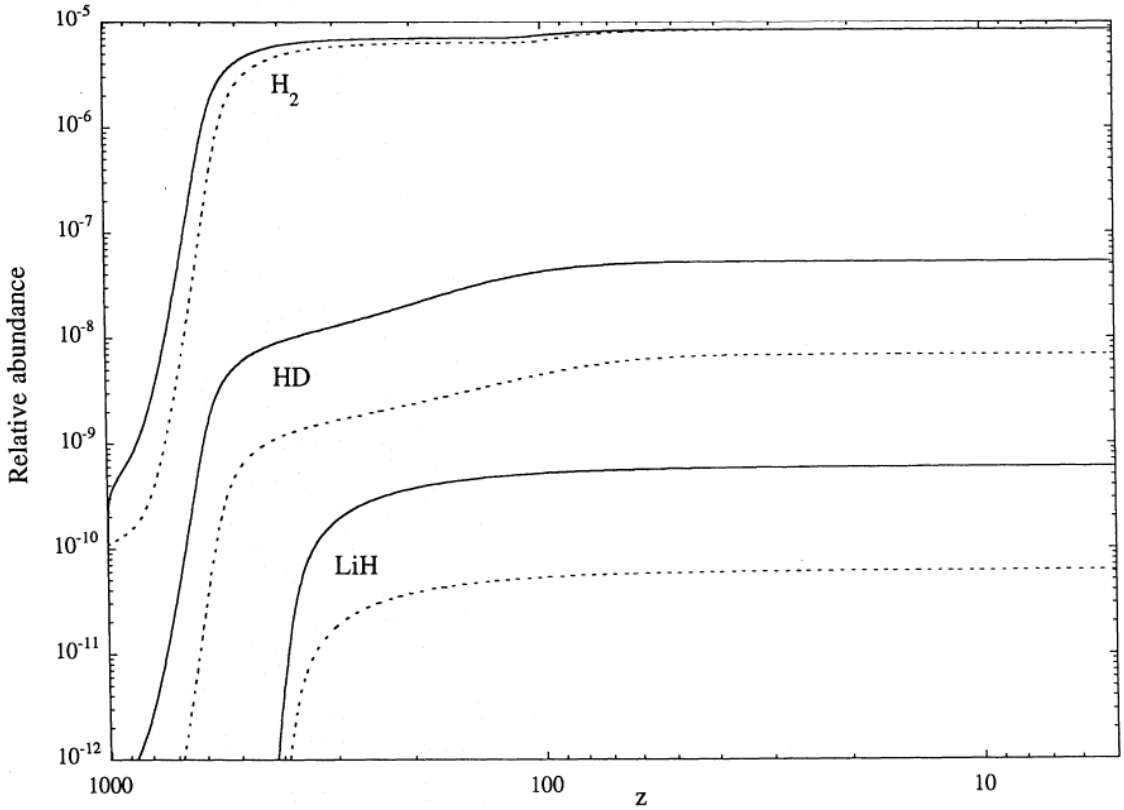
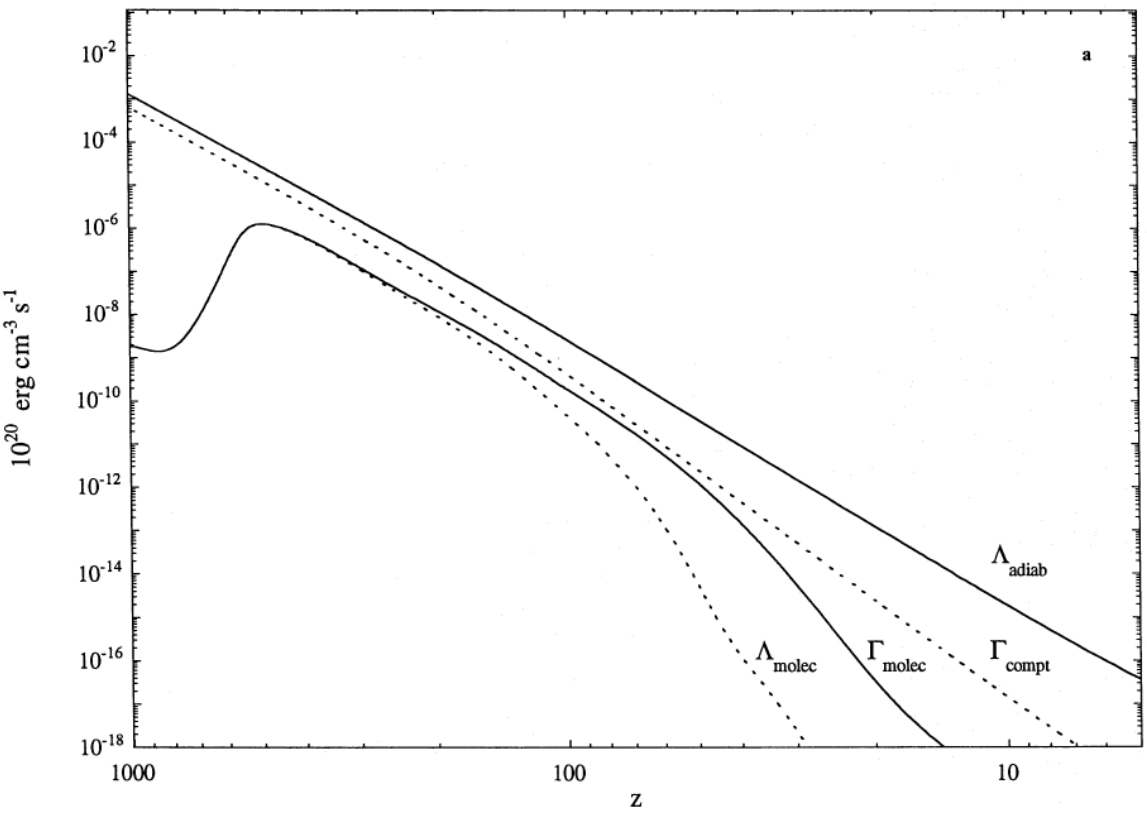


Fig. 1. Comparison of the fractional abundances of H_2, HD, LiH for models A (dotted line) and B (continuous line). After a transient growth H_2 abundance becomes almost constant around $8\ 10^{-6}$. The two plateau in H_2 and HD curves correspond to successive saturations of the two different formation channels. Note the importance of photodissociation of LiH for $z \geqq 400$

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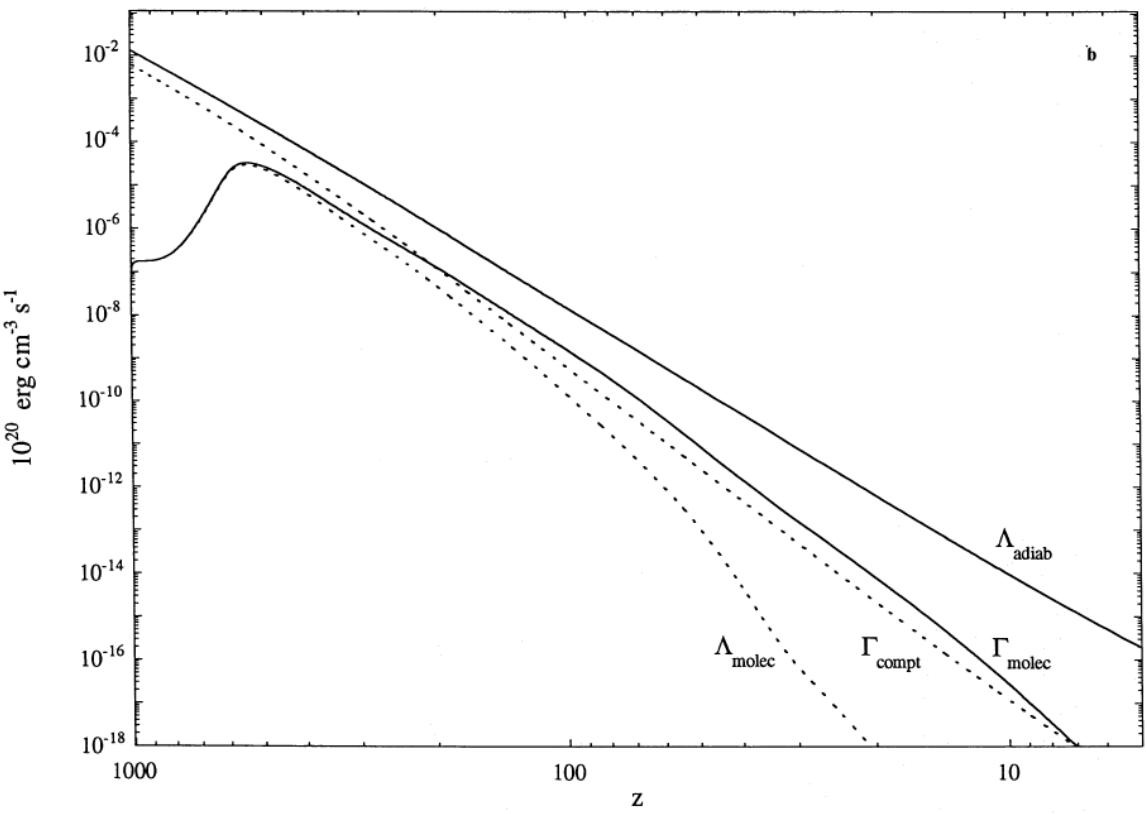


Fig. 2a and b. Cooling and heating evolution rates of models A (**a**) and B (**b**). The upper continuous curve represents the adiabatic cooling (Λ_{adiab}), the Γ_{Compt} curve is the Compton heating, and the two lower curves give the molecular heating (Γ_{mol}) and cooling Λ_{mol})

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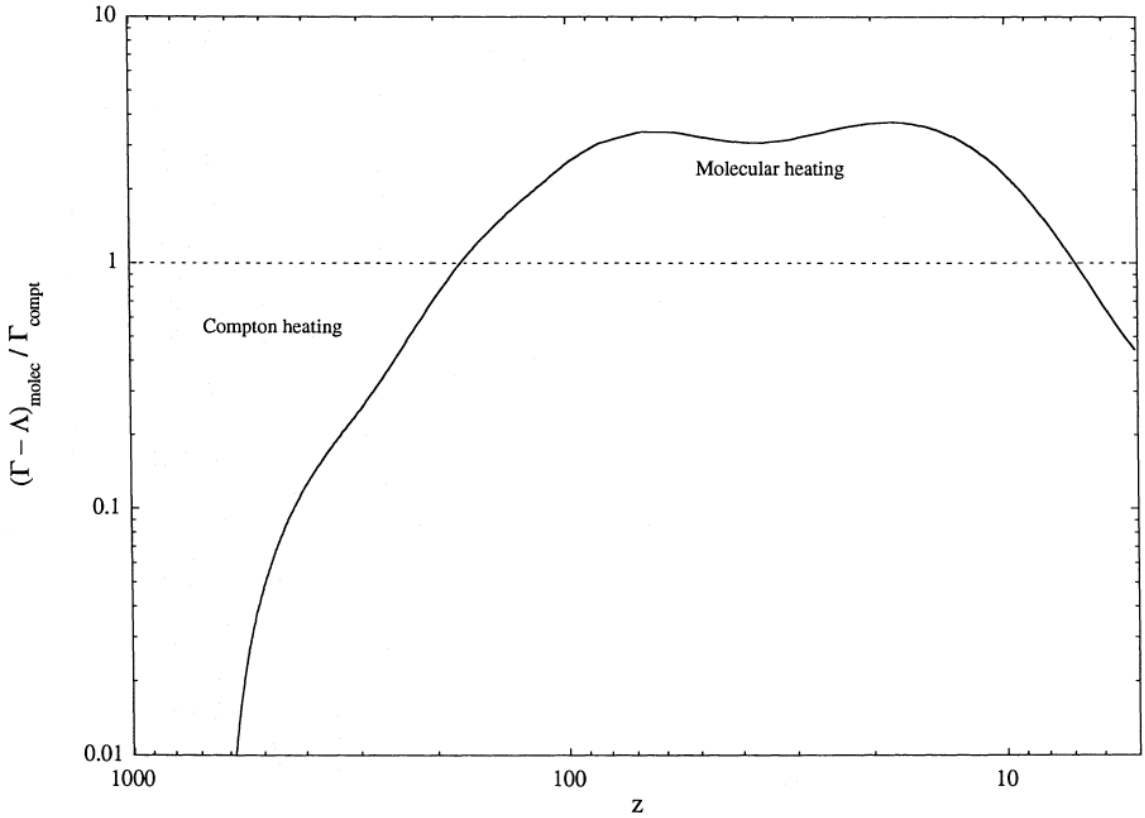


Fig. 3. Comparison between molecular and Compton heating for model B. The curve represents the ratio $\frac{\Gamma_{mol} - \Lambda_{mol}}{\Gamma_{Compt}}$ of the molecular heat-loss function to the Compton heating. Molecular heating become greater than Compton heating for $z \lesssim 180$

5. Results and discussion

We have considered two different models A and B (Table 1) whose parameters are Ω_0 and Ω_b (the total and baryonic density parameters) and the relative elemental abundance of deuterium and lithium. In these two models, $H_0 = 50\,km\,s^{-1}\,Mpc^{-1}$.

Model A corresponds to the standard model of nucleosynthesis of Wagoner et al. (1967) and Yang et al. (1984). The relative elemental abundances of model B are given by the inhomogeneous primordial nucleosynthesis models (Applegate & Hogan 1985, Sale & Mathews 1986, Mathews et al. 1990). In the calculations presented below the integration starts at $z = 1000$ with an electron abundance of 10^{-1}, we take the following initial values $H^+/H = D^+/D = 1/9$ (see Lepp & Shull 1984 for a detailed study of the recombination period).

Relative abundances of H_2, HD and LiH are shown on Fig. 1 for both models. H_2 abundances rises through two processes. The first step corresponds to the H_2^+ channel and the second one to the H^- channel. The HD abundance rises also in two steps with a transition around $z = 100$, the D^+ channel (reaction 15), and the HD^+ channel (reaction 18). LiH formation through radiative association becomes effective when the photodestruction decreases ($z \lesssim 400$). These results are similar to those of Lepp & Shull (1984) and Black (1988).

In Fig. 2a, molecular heating and cooling are compared with the contribution due to the Compton heating. The main source of heating is due to the Compton process. In model B (Fig. 2b), as expected, the molecular contribution is more important than in model A and the molecular heating dominates the Compton heating when $z \lesssim 180$ (see Fig. 3).

The evolution of the mean temperature of the matter is shown in Fig. 4 for model A and B. The evolution of matter temperature is shown without and with the molecular contributions, the evolution of radiation temperature is also plotted. The ionisation fraction is higher when the baryon density is low (model A see Fig. 1), then the Compton scattering is important and the molecular contribution remains small.

In spite of the small influence of the molecular process on the evolution of the mean temperature of matter, a question may be asked: could the molecular radiation contributes to distorsions of the cosmological background? Molecular cooling function corresponds to the emission of photons which can add up to the background radiation, whereas molecular heating corresponds to an absorption of the photons of background radiation. Fig. 5 shows the various contributions of the different molecules to the molecular heating (Fig. 5a) and cooling (Fig. 5b). These processes are dominated successively by H_2, HD and LiH. The contribution of H_2 dominates first. For $z < 150$ collisonal excitation in the excited rotational states of H_2 becomes negligible ($T_m < 300\,K$). HD heating becomes efficient when $z \lesssim 50$ and HD cooling when redshifts ranging between 120 and 30, the contribution of LiH is effective when $z \lesssim 30$ and remains small.

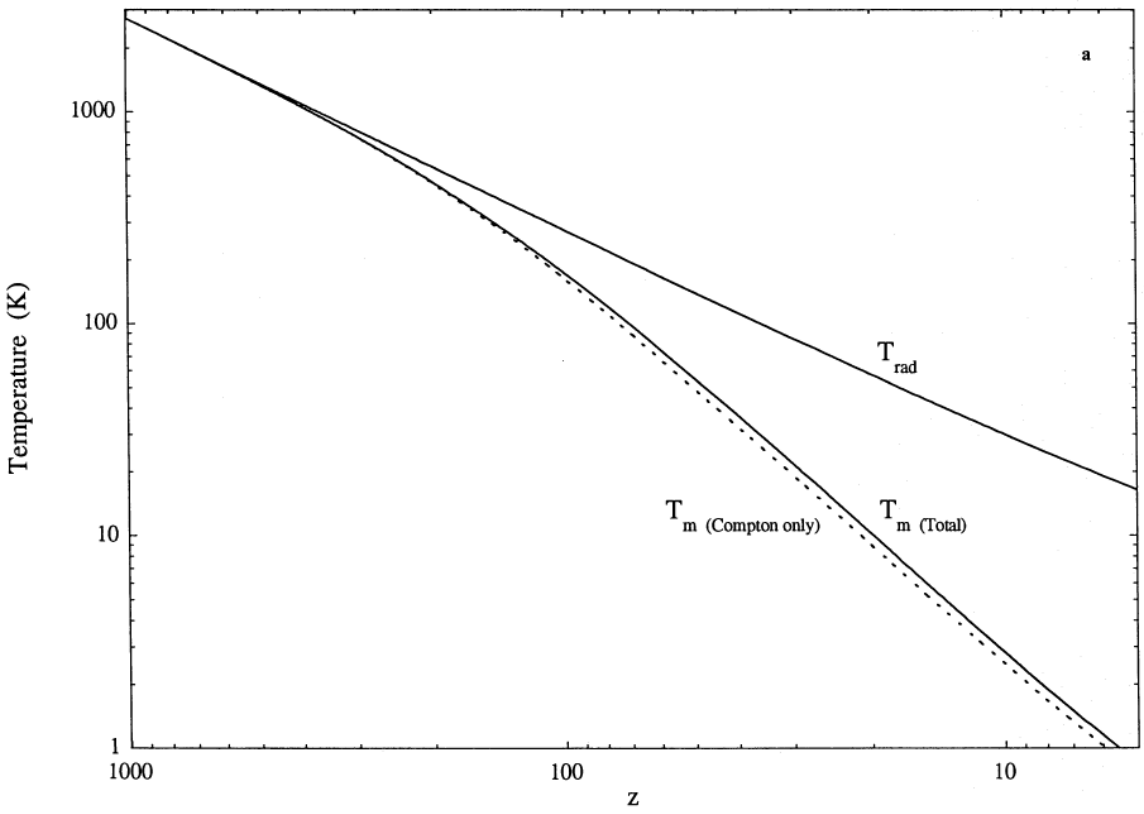
Thus, for transitions occurring at z, the flux ϕ^{ij} observed at $z = 0$ at a frequency ν_{ij} is:

$$\nu^{ij} = \frac{\nu_0^{ij}}{1+z}, \tag{11}$$

$$\phi^{ij} = \frac{c}{4} \frac{\Gamma_{mol}^{ij} - \Lambda_{mol}^{ij}}{\nu_0^{ij}} \frac{1}{H_0} \frac{1}{(1+z)^4 \sqrt{1+\Omega_0 z}}, \tag{12}$$

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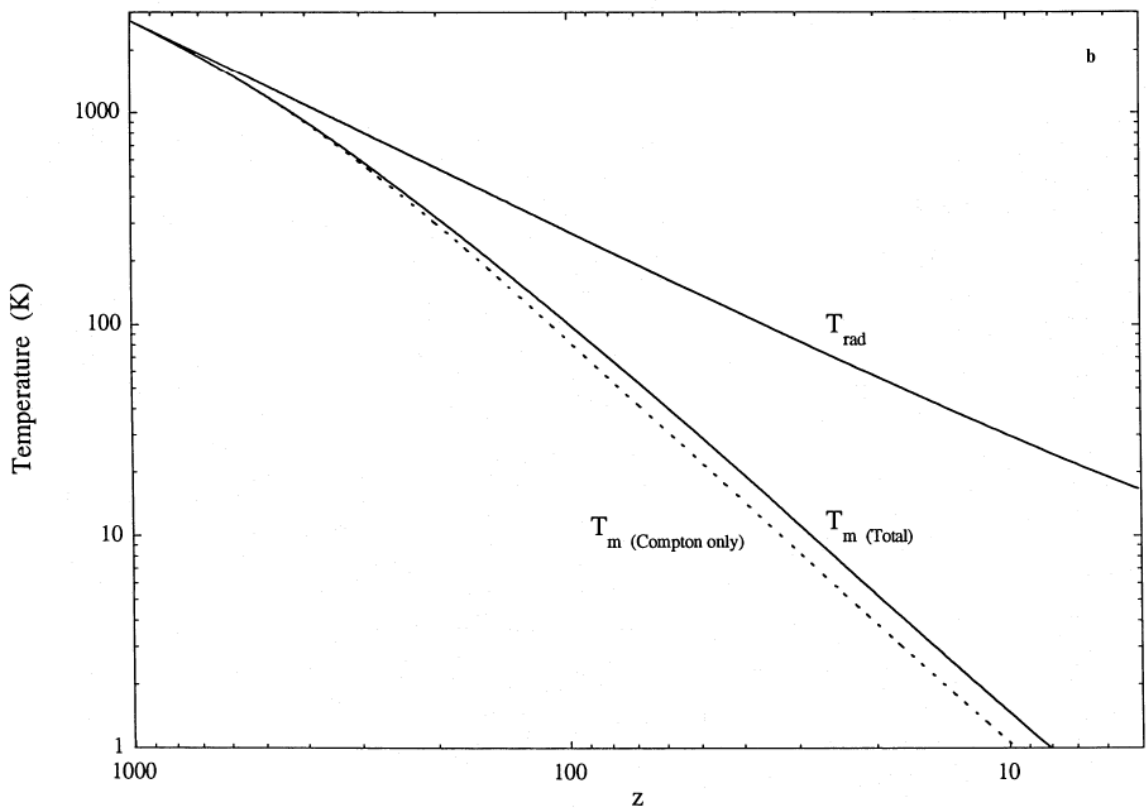


Fig. 4a and b. Thermal history of models A and B. The three curves represent the evolution of the radiation temperature (T_r) and the matter temperature without molecular contribution ($T_{m(Compton\ only)}$) and with molecular contribution ($T_{m(total)}$). The molecular contribution is greater in model B

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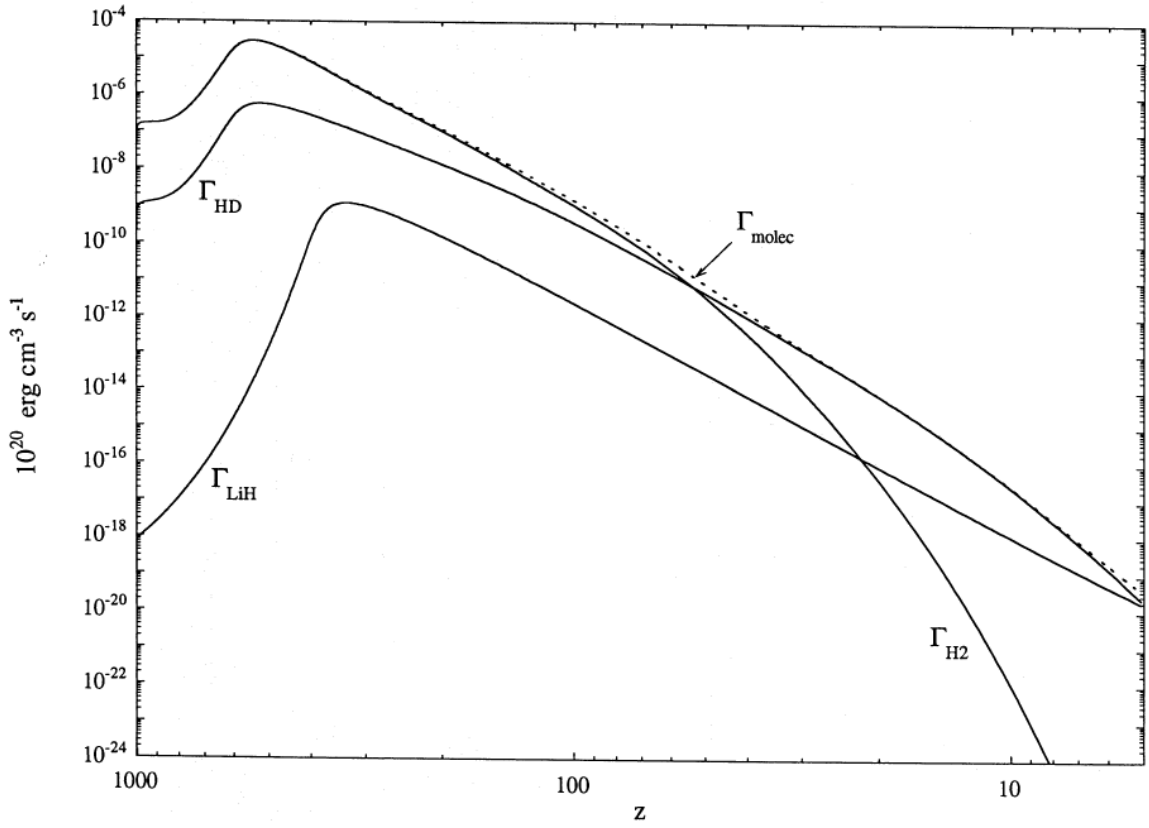


Fig. 5a. Detailed evolution of the molecular heating (Γ_{mol}) in model B. Γ_{H_2}, Γ_{HD} and Γ_{LiH} represent respectively the contribution of the H_2, HD and LiH molecules

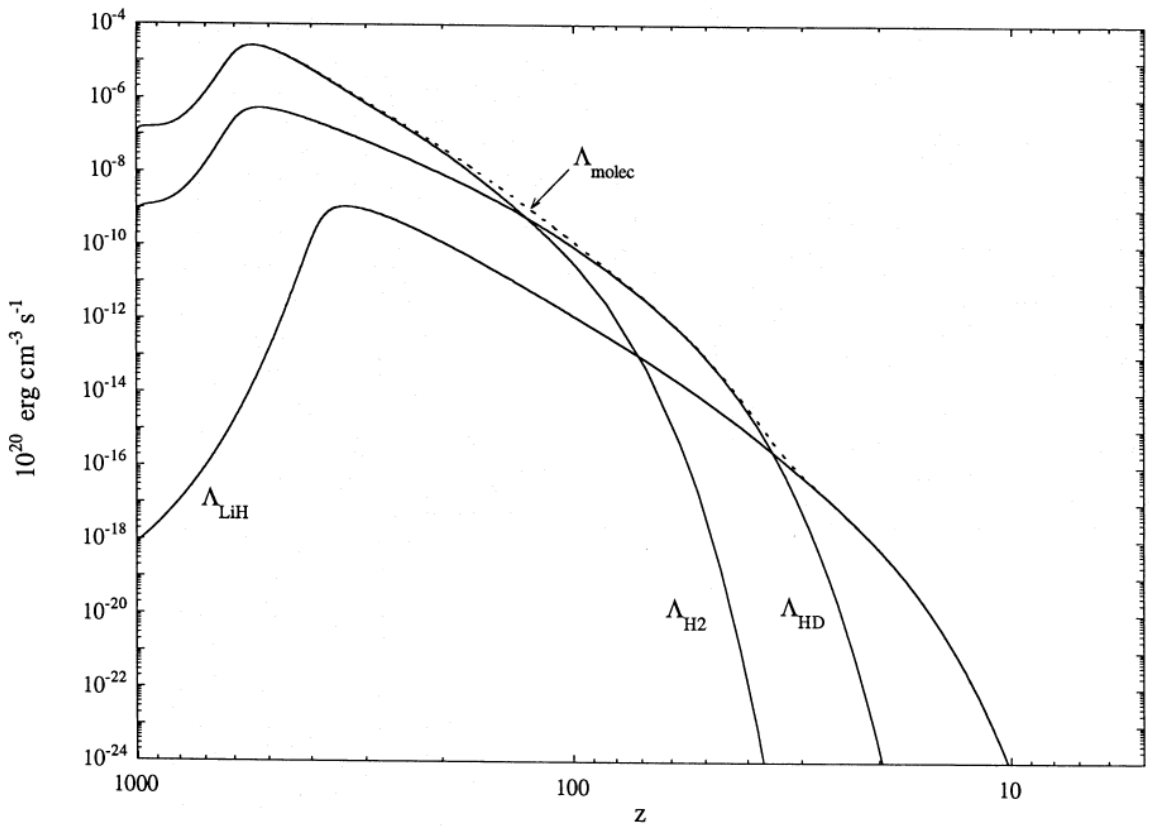


Fig. 5b. Detailed evolution of the molecular cooling (Λ_{mol}) in model B. Λ_{H_2}, Λ_{HD} and Λ_{LiH} represent respectively the contribution of the H_2, HD and LiH molecules

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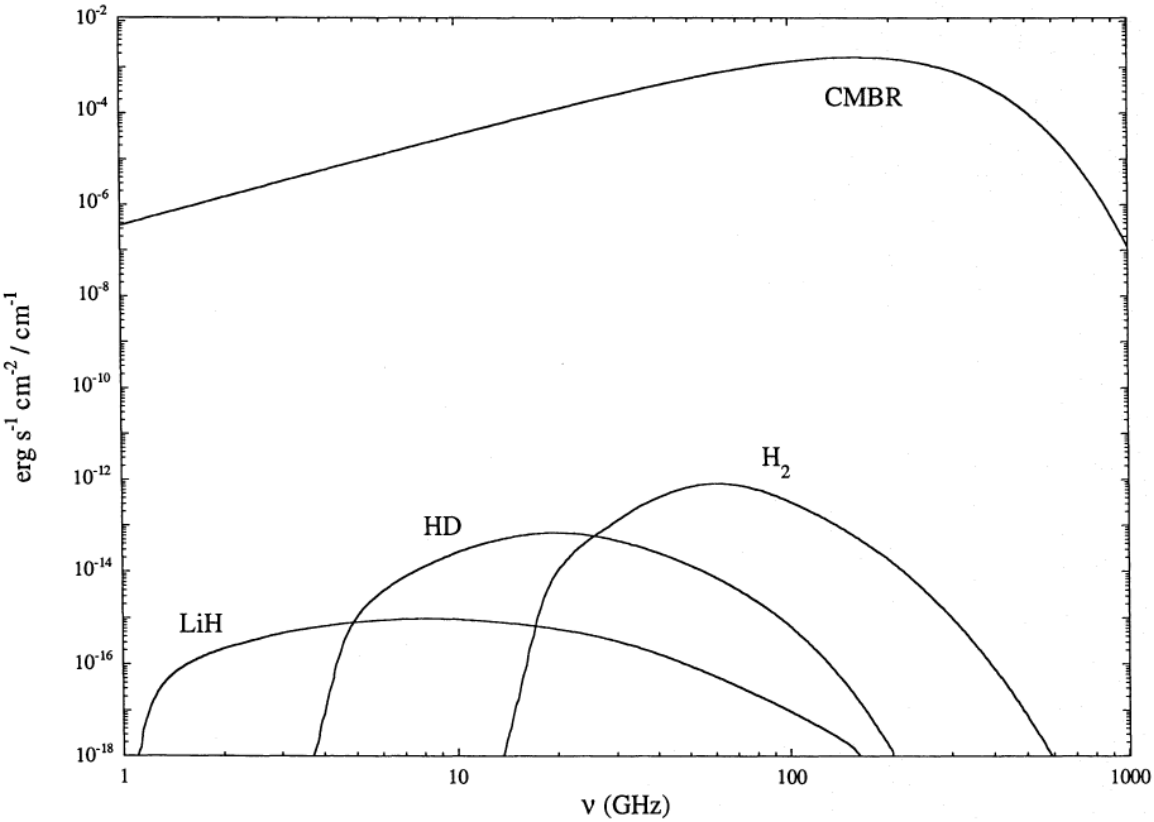


Fig. 6. Comparison between the integrated fluxes of H_2, HD and LiH at $z = 0$ and the CMBR

where ν_0^{ij} is the rest frequency of the transition i, j. ϕ_{ij} is then summed over z.

Figure 6 compares the theoretical flux of each molecule (model B), and the CMBR measured by the Far InfraRed Absolute Spectrophotometer (FIRAS) on the COBE satellite (Mather et al. 1990) with a spectral resolution of 1 cm^{-1}. This instrument shows that the background radiation has a blackbody spectrum with a temperature of 2.735 K between 30 and 600 GHz. The main molecular contribution to the total flux is due to H_2, which has a maximum nine orders of magnitude below CMBR in model B. Thus, molecular radiative processes cannot be detected by FIRAS for a homogeneous distribution.

In conclusion, the primordial molecules seem to play a moderate role in the thermal history of the Universe. Nevertheless, molecules could play a more decisive role in the galaxy formation, since molecular cooling can trigger thermal instability prior to gravitational instability. This scenario has been proposed by Fall and Rees (1985). Recently the Differential Microwave Radiometer instrument (DMR) on COBE (Smoot et al. 1992) have measured the large-angular-scale anisotropy of the CMBR and the maximum fluctuation. The numerical procedures considered in the present paper will be used to study the formation of primordial molecular clouds and the possible occurrence of thermal instability, since molecular abundances and thermal fluxes are time dependent.

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Appendix: table of chemical reactions

The reaction rates are $k = \gamma \left(\frac{T}{300}\right)^{\alpha} \exp\left(\frac{-\beta}{T}\right)$, where γ is in $cm^3\,s^{-1}$, β in Kelvin. The enthalpy ΔH is in eV

Reactions		γ	α	β	ΔH	References
(1)	$H + e \mapsto H^- + h\nu$	$3\ 10^{-16}$	1	0		Prasad & Huntress (1980)
(2)	$H^- + H \mapsto H_2 + e$	$1.5\ 10^{-9}$	0	0	3.71	Launay et al. (1991)
(3)	$H^+ + H \mapsto H_2^+ + h\nu$	$1.8\ 10^{-18}$	1.5	0		Dalgarno & Lepp (1987)
(4)	$H_2^+ + H \mapsto H_2 + H^+$	$6.4\ 10^{-10}$	0	0	1.81	Karpas et al. (1979)
(5)	$H_2 + h\nu \mapsto H + H$	See Sect. 1				Latter & Black (1991)
(6)	$H^- + h\nu \mapsto H + e$	See Sect. 1				Wishart (1979)
(7)	$H_2^+ + h\nu \mapsto H^+ + H$	See Sect. 1				Von Busch & Dunn (1972)
(8)	$H^+ + H^- \mapsto H + H$	$2.3\ 10^{-7}$	-0.5	0	12.84	Prasad & Huntress (1980)
(9)	$H^+ + H^- \mapsto H_2^+ + e$	$8.83\ 10^{-15}$	-0.32	0	1.9	Poulaerts et al. (1978)
(10)	$H_2^+ + H^- \mapsto H_2 + H$	$2.3\ 10^{-7}$	-0.5	0	14.66	Prasad & Huntress (1980)
(11)	$H^+ + e \mapsto H + h\nu$	$3.5\ 10^{-12}$	-0.70	0		Prasad & Huntress (1980)
(12)	$H_2^+ + e \mapsto H + H$	$1.68\ 10^{-8}$	-0.29	0	10.95	Nakashima et al. (1987)
(13)	$H^+ + D \mapsto D^+ + H$	10^{-9}	0	41	$-$ 0.01	Watson (1976)
(14)	$H + D \mapsto HD + h\nu$	10^{-25}	0	0		Lepp & Shull (1984)
(15)	$D^+ + H_2 \mapsto H^+ + HD$	$1.7\ 10^{-9}$	0	0	0.06	Smith et al. (1982)
(16)	$D^+ + H \mapsto HD^+ + h\nu$	$1.8\ 10^{-18}$	1.5	0		Dalgarno & Lepp (1987)
(17)	$H^+ + D \mapsto HD^+ + h\nu$	$1.8\ 10^{-18}$	1.5	0		Dalgarno & Lepp (1987)
(18)	$HD^+ + H \mapsto H^+ + HD$	$6.4\ 10^{-10}$	0	0	1.85	Karpas et al. (1979)
(19)	$HD + h\nu \mapsto H + D$	See Sect. 1				Latter & Black (1991)
(20)	$HD^+ + h\nu \mapsto H^+ + D$	See Sect. 1				Von Busch & Dunn (1972)
(21)	$HD^+ + h\nu \mapsto D^+ + H$	See Sect. 1				Von Busch & Dunn (1972)
(22)	$D^+ + e \mapsto D + h\nu$	$3.5\ 10^{-12}$	-0.70	0		Prasad & Huntress (1980)
(23)	$HD^+ + e \mapsto D + H$	$1.39\ 10^{-8}$	-0.29	0	10.93	Nakashima et al. (1987)
(24)	$D^+ + H \mapsto H^+ + D$	10^{-9}	0	0	0.01	Watson (1976)
(25)	$H^+ + HD \mapsto D^+ + H_2$	$1.7\ 10^{-10}$	0	462	$-$ 0.06	Smith et al. (1982)
(26)	$H_2^+ + D \mapsto D^+ + H_2$	$6.4\ 10^{-10}$	0	0	1.81	Karpas et al. (1979)
(27)	$D + H_2 \mapsto HD + H$	$1.09\ 10^{-18}$	0	0	0.05	Zhang & Miller (1989)
(28)	$Li + H \mapsto LiH + h\nu$	10^{-17}	0	0		Dalgarno & Lepp (1987)
(29)	$LiH + h\nu \mapsto Li + H$	See Sect. 1				Kirby & Dalgarno (1978)

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